PHASETRACKER

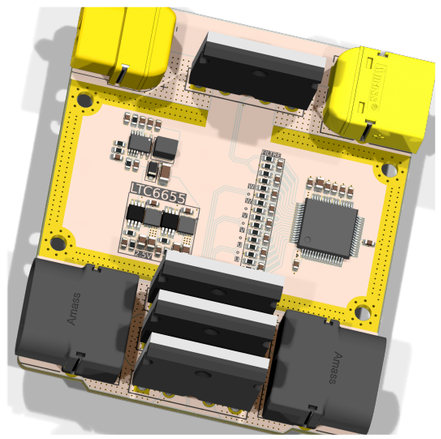
2026-10-18

Variant:

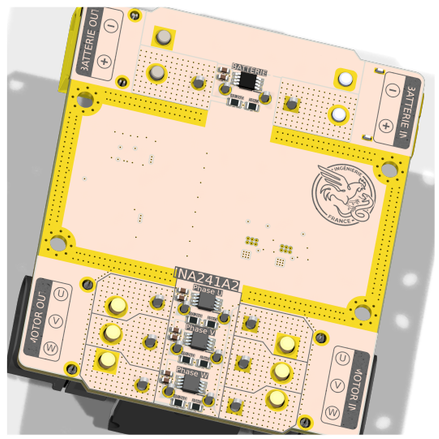
Rev + (Unreleased)

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4	AD7606C-18_ADC	14	.....	24	.....	34	.....
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6	Power_Supply	16	.....	26	.....	36	.....
7	Connectors	17	.....	27	.....	37	.....
8	Holes, Fiducials	18	.....	28	.....	38	.....
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TOP VIEW



BOTTOM VIEW



NOTES

3-phase current and voltage sensing: INA241A2 + AD7606C-18

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 18-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Current: 2.5 V - 20 mV/A x I(J1->J2).
Voltage: 0.198 V/V.

DESIGN NOTE:
No Power-Good: rails within 0.1 % after 71 ms.

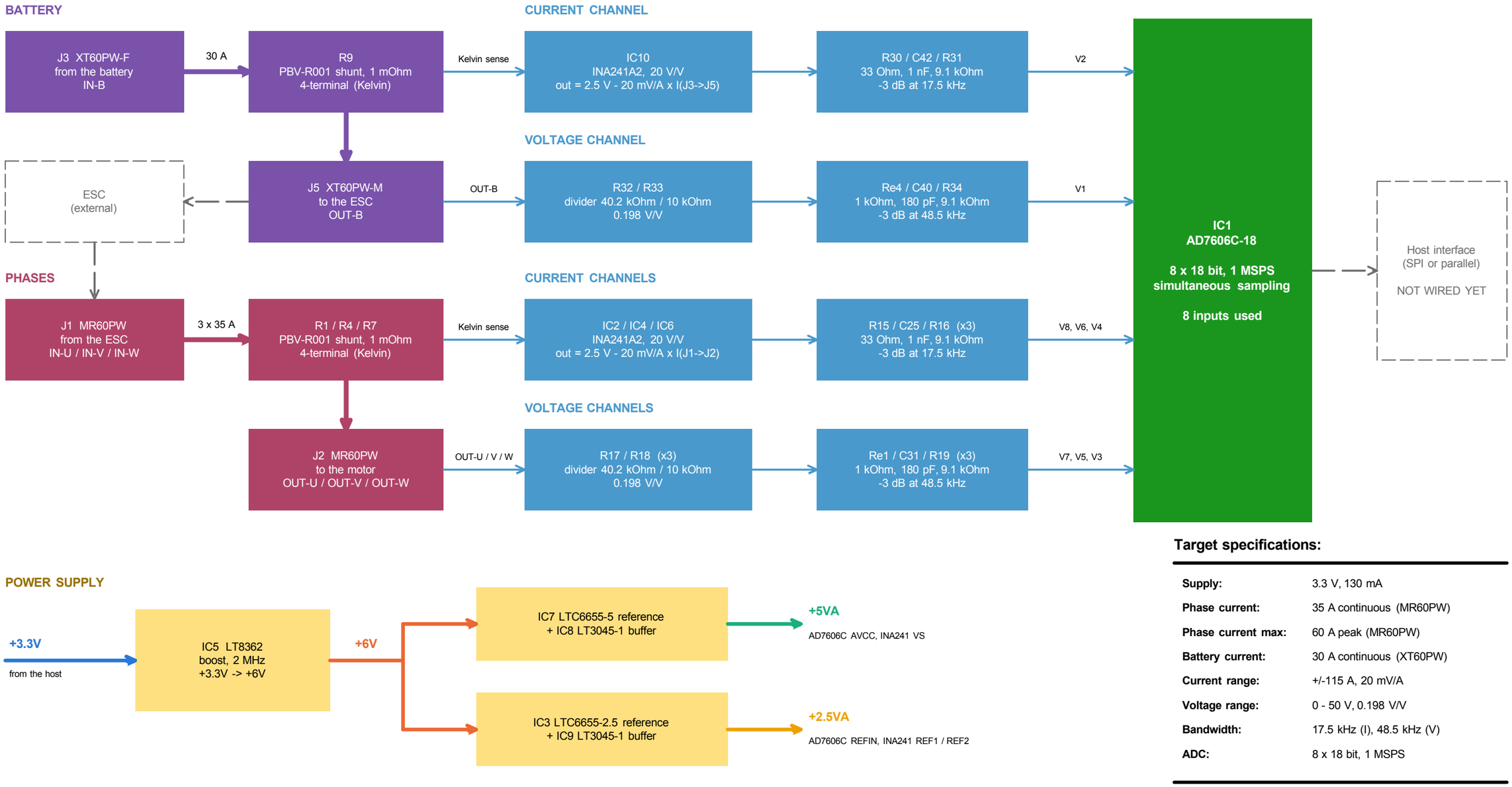
DESIGN NOTE:
AD7606C digital pins and +3.3V input not wired yet.

DESIGN NOTE:
+2.5VA: R12 = 49.9k gives 2.55 V, 24.9k gives 2.50 V. See README.

LAYOUT NOTE:
Kelvin-route the shunt sense pads to each INA241.

	Comments:	Company: OPEN THRUST LAB	Variant:	Git Hash: 7703dc5
		Board Name: PHASETRACKER	Project Name: OPEN_PHASE	
	Sheet Title: OPEN_PHASE - PHASETRACKER	File Name: OPEN_PHASE.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-06 / Revision: + (Unreleased)
	Sheet Path: /		Reviewer:	Size: A3 / Sheet: 1 of 10

[2] Block Diagram



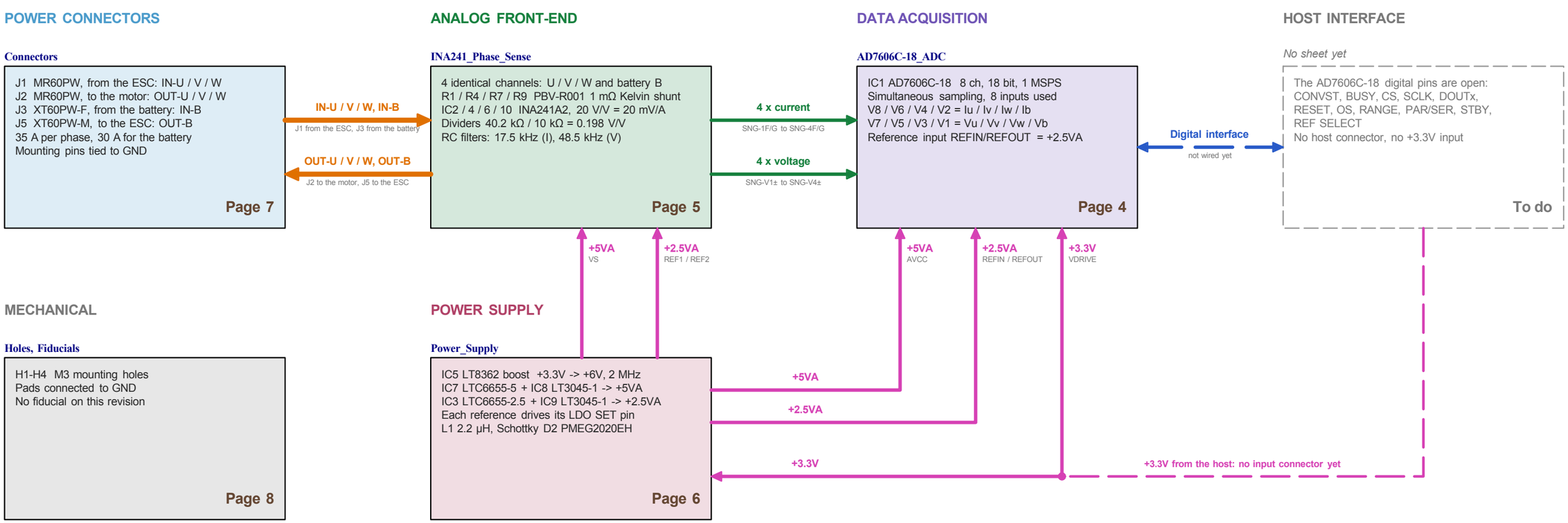
Target specifications:

Supply:	3.3 V, 130 mA
Phase current:	35 A continuous (MR60PW)
Phase current max:	60 A peak (MR60PW)
Battery current:	30 A continuous (XT60PW)
Current range:	+/-115 A, 20 mV/A
Voltage range:	0 - 50 V, 0.198 V/V
Bandwidth:	17.5 kHz (I), 48.5 kHz (V)
ADC:	8 x 18 bit, 1 MSPS

Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 7703dc5
	Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-11	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 10

[3] Project Architecture

Sheet hierarchy and signal flow. Click a block or its page number to open the sheet.



LEGEND

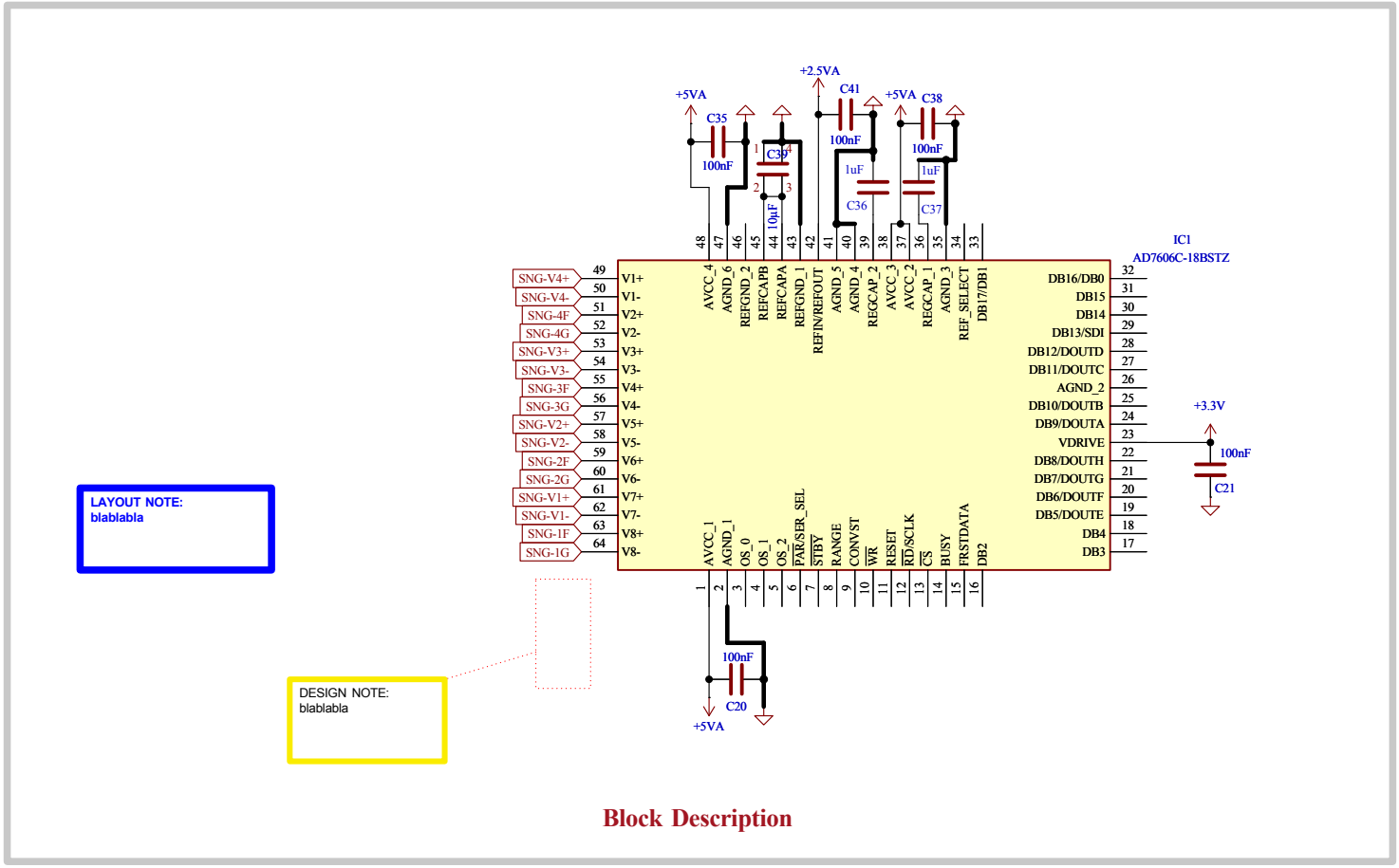
- Power conductor: phases 35 A, battery 30 A
- Power rail
- Analog signal, differential pairs
- Digital interface, not wired yet
- Supply input, no connector yet

DOCUMENT INDEX

[1] Title Page
[2] Block Diagram
[3] Project Architecture
[4] AD7606C-18 ADC
[5] INA241 Phase Sensing
[6] Power Supply - Analog Rails
[7] Connectors
[8] Holes, Fiducials
[9] Power Sequencing
[10] Revision History

Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 7703dc5
	Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-11	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 10

[4] AD7606C-18
- ADC



Block Description

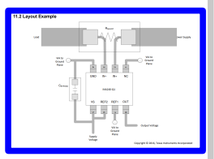
Comments: AN5346; STM32G474 Datasheet p.81; J. Pieper ADC investigation	Company: OPEN THRUST LAB		Variant:	Git Hash: 7703dc5
	Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
Sheet Title: AD7606C-18 - ADC	File Name: AD7606C-18 - ADC.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-05	Revision: + (Unreleased)
Sheet Path: /Project Architecture/AD7606C-18_ADC/		Reviewer:	Size: A4	Sheet: 4 of 10

[5] INA241 - Phase Sensing

The INA240 device is a voltage-output, current-sense amplifier with enhanced PWM rejection that can sense drops across shunt resistors over a wide common-mode voltage range from -4 V to 80 V, independent of the supply voltage. The negative common-mode voltage allows the device to operate below ground, accommodating the flyback period of typical solenoid applications. Enhanced PWM rejection provides high levels of suppression for large common-mode transients (ΔV/Δt) in systems that use pulse width modulation (PWM) signals (such as motor drives and solenoid control systems). This feature allows for accurate current measurements without large transients and associated recovery ripple on the output voltage.
This device operates from a single 2.7-V to 5.5-V power supply, drawing a maximum of 2.4 mA of supply current. Four fixed gains are available: 20 V/V, 50 V/V, 100 V/V, and 200 V/V. The low offset of the zero-drift architecture enables current sensing with maximum drops across the shunt as low as 10-mV full-scale. All versions are specified over the extended operating temperature range (-40°C to +125°C), and are offered in an 8-pin TSSOP and 8-pin SOIC packages.

DESIGN NOTE:
Gain A1->A5 (INA241)
INA241A1 : 10V/V
INA241A2 : 20V/V (fitted)
INA241A3 : 50V/V
INA241A4 : 100V/V
INA241A5 : 200V/V

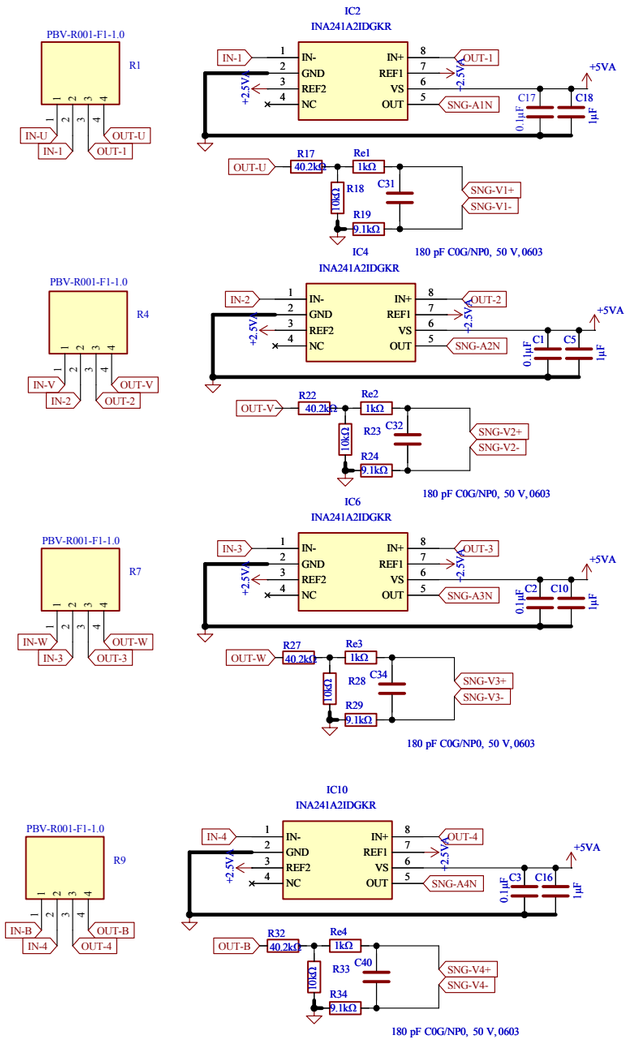
DESIGN NOTE:
Capacitor Découplage
0603
Ceramique X7R
>16V



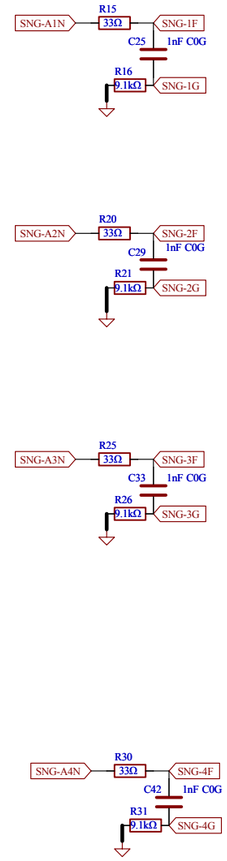
Description

Vout≈Vref+(Gain)×(I×Rshunt)

Shouse your INA240A...



INA241A + Shunt



FILTRE

Test Point

Comments: **References:** Flexible I/O worked examples; Flexible I/O source configuration	Company: **OPEN THRUST LAB**		Variant:	Git Hash: 7703dc5
	Board Name: **PHASETRACKER**		Project Name: **OPEN_PHASE**	
Sheet Title: INA241 - Phase Sensing	File Name: INA241 - Phase Sensing.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-05	Revision: + (Unreleased)
Sheet Path: /Project Architecture/INA241_Phase_Sense/		Reviewer:	Size: **A3**	Sheet: **5** of **10**

[6] Power Supply
- Analog Rails

DESIGN NOTE:
Gain A1->A5 (INA241)
INA241A1 : 10V/V
INA241A2 : 20V/V
(fitted)
INA241A3 : 50V/V
INA241A4 : 100V/V
INA241A5 : 200V/V

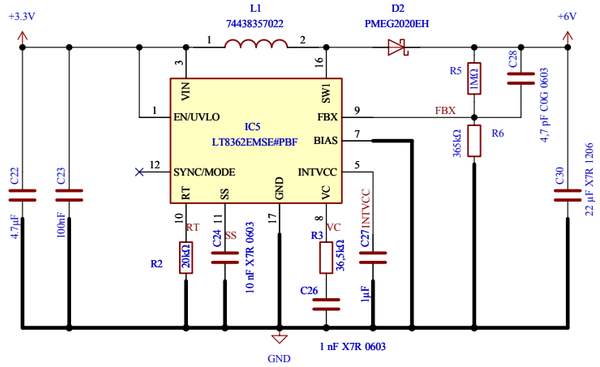
DESIGN NOTE:
Capacitor Découplage
0603
Ceramique X7R
>16V

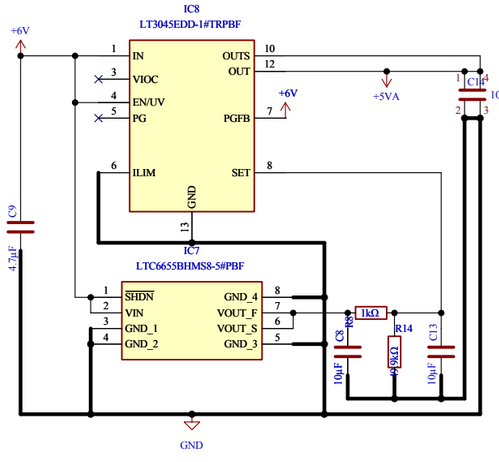
LAYOUT NOTE:
blabla

Description

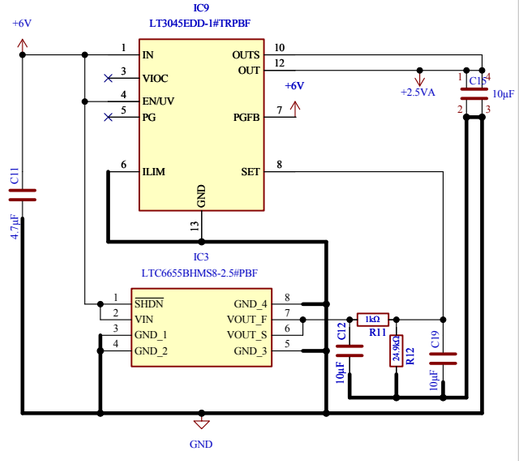
Vout≈Vref+(Gain)×(I×Rshunt)

Shouse your INA240A...





5VA précision Alimentation



2.5VA précision Alimentation

Don't panic

it's just a regular capacitor with a slightly modified footprint to better cover the 1 MHz and higher bands.

Non-critical detail.

Test Point

Comments: **References:** Flexible I/O worked examples Flexible I/O source configuration	Company: **OPEN THRUST LAB**		Variant:	Git Hash: 7703dc5
	Board Name: **PHASETRACKER**		Project Name: **OPEN_PHASE**	
Sheet Title: Power Supply - Analog Rails	File Name: Power Supply - Analog Rails.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-06	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power_Supply/		Reviewer:	Size: **A3**	Sheet: **6** of **10**

[7] Connectors

MR60PW stat :

- **Current: 35 A max (with specified temperature rise < 85°C)**
- **Voltage: 600 V DC**
- **Insulation resistance: ≥ 2000 MΩ**
- **Contact resistance: ≤ 1.0 mΩ**
- **Mating cycles (lifetime): 100 cycles**
- **Salt spray: 48 h**
- **Ingress protection rating: IP40**
- **Operating temperature: -20°C to 120°C**
- **Flammability rating: UL94 V-0 (0.8 mm wall thickness)**

MR60PW

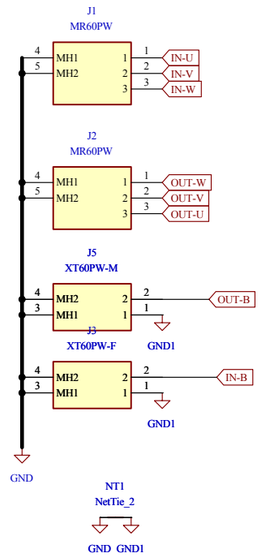
DESIGN NOTE:
Gain A1->A5 (INA241)
INA241A1 : 10V/V
INA241A2 : 20V/V
(fitted)
INA241A3 : 50V/V
INA241A4 : 100V/V
INA241A5 : 200V/V

DESIGN NOTE:
Capacitor Découplage
0603
Ceramique X7R
>16V

LAYOUT NOTE:
blabla

Vout≈Vref+(Gain)×(I×Rshunt)

Shouse your INA240A...



CONNECTEUR

FILTRE

Test Point

Comments: **References:** Flexible I/O worked examples; Flexible I/O source configuration	Company: **OPEN THRUST LAB**		Variant:	Git Hash: 7703dc5
	Board Name: **PHASETRACKER**		Project Name: **OPEN_PHASE**	
Sheet Title: Connectors	File Name: Connectors.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-05	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: **A3**	Sheet: **7** of **10**

[8] Holes - Fiducials

DESIGN NOTE:
Gain A1->A5 (INA241)
INA241A1 : 10V/V
INA241A2 : 20V/V
(fitted)
INA241A3 : 50V/V
INA241A4 : 100V/V
INA241A5 : 200V/V

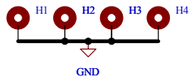
DESIGN NOTE:
Capacitor Découplage
0603
Ceramique X7R
>16V

LAYOUT NOTE:
blabla

Description

Vout≈Vref+(Gain)×(I×Rshunt)

Shouse your INA240A...



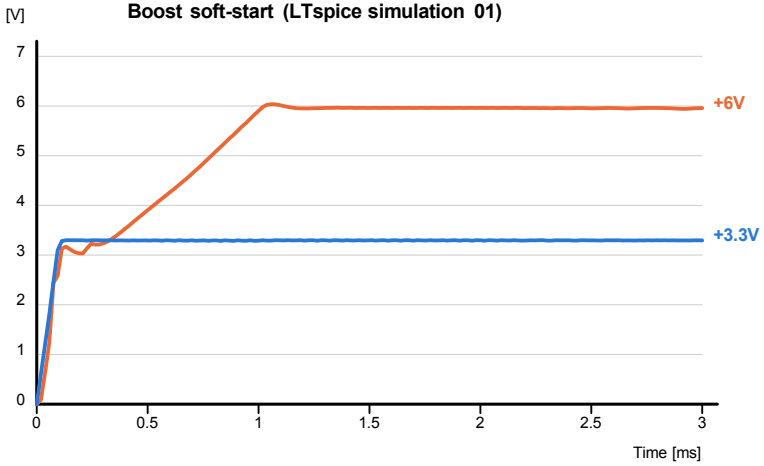
INA240 + Shunt

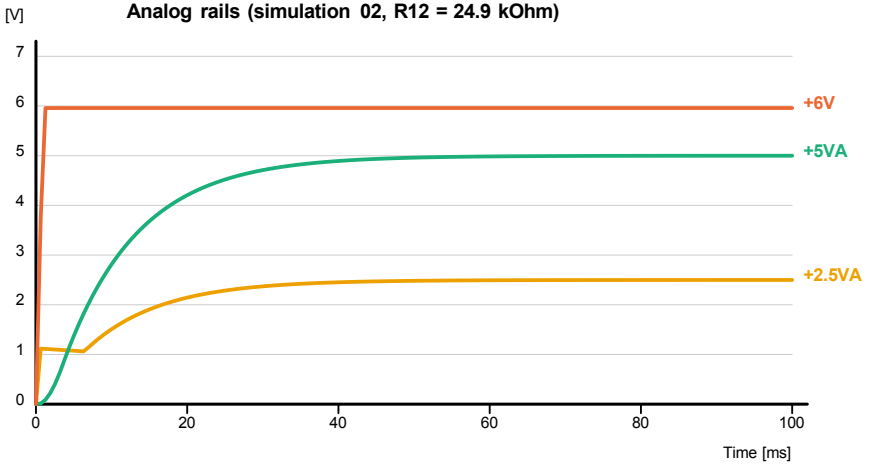
FILTRE

Test Point

Comments: **References:** Flexible I/O worked examples; Flexible I/O source configuration	Company: **OPEN THRUST LAB**		Variant:	Git Hash: 7703dc5
	Board Name: **PHASETRACKER**		Project Name: **OPEN_PHASE**	
Sheet Title: Holes - Fiducials	File Name: Holes - Fiducials.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-05	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Holes, Fiducials/		Reviewer:	Size: **A3**	Sheet: **8** of **10**

[9] Power - Sequencing





NAME	SOURCE	LEVEL	TIMING (after +3.3V is applied)
+3.3V	Host (no connector on the board yet)	3.3 V	t = 0
+6V	IC5 LT8362 boost, 2 MHz, soft-start C24 = 10 nF	5.96 V	90 % after 0.9 ms
+5VA	IC7 LTC6655-5 -> R8 -> SET of IC8 LT3045-1	5.000 V	1 % after 48 ms, 0.1 % after 71 ms (tau = 9.8 ms)
+2.5VA	IC3 LTC6655-2.5 -> R11 -> SET of IC9 LT3045-1	2.500 V (*)	same time constant as +5VA

(*) As built R12 = 49.9 kOhm gives 2.549 V. The curve and the level are given for R12 = 24.9 kOhm. See README, DESIGN REVIEW 1.

No Power-Good signal (PGFB tied to IN): wait at least 100 ms after power-up before the first conversion.

Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 7703dc5
	Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-06	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 9 of 10

[10] Revision History

Version Unreleased

Added

- 3x PBV 1 mOhm shunt + INA241A2 (20 V/V)
- 3x phase voltage divider 40.2k / 10k
- AD7606C-18 ADC, 6 inputs used
- +3.3V -> LT8362 6 V -> LT3045-1 +5VA / +2.5VA
- Self-contained libs and 3D models (lib/)
- LTspice simulations (Simulation/LTspice)
- Predicted measurement quality (sim 07, 08)
- openEMS shunt inductance model (Simulation/openEMS)
- Thermal / current density model (Simulation/Thermal)
- 3D animation (Images/OPEN_PHASE.mp4)

Changed

- Renamed to OPEN_PHASE, KiCad 10
- Sheets named after their function
- IC3: LTC6655LN-2.5 -> LTC6655-2.5 (pin 6 = VOUT_S)
- R12: 49.9k -> 24.9k (+2.5VA = 2.500 V)
- L1: 744373965022 -> WE-MAPI 4030 74438357022 (4.1 x 4.1 mm)
- Solder mask white, silkscreen black

Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 7703dc5
	Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-05	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 10 of 10